

|          |                                                                                                                                                                                                                                   |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)(i)  | $(26 \div 288) = 0.090$ ;                                                                                                                                                                                                         | <b>1</b> |
| 2(a)(ii) | the data are, categoric / nominal ;                                                                                                                                                                                               | <b>1</b> |
| 2(b)(i)  | <p>number of people with (blood) cancer increases with (mother's) exposure to radiation ;</p> <p>idea of a linear or proportional relationship ;</p> <p>no exposure to radiation still shows (blood) cancer developing / AW ;</p> | <b>3</b> |
| 2(b)(ii) | <i>idea that</i> radiation exposure is linked to, mutation / DNA damage, of stem cells (in bone marrow) ;                                                                                                                         | <b>1</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(c)     | <p>any <b>three</b> from:</p> <p><i>increases validity:</i></p> <ol style="list-style-type: none"> <li>1 <i>idea that</i> a large number in the study ;</li> <li>2 <i>idea that</i> data was collected over, a long period of time / 61 years / 56 years ;</li> <li>3 <i>idea that</i> cohorts, all from / stayed, in the same area (so exposure similar) ;</li> <li>4 initial source of radiation the same ;</li> </ol> <p><i>decreases validity:</i></p> <ol style="list-style-type: none"> <li>5 <i>idea that</i> group 1 people were born over longer time (40 years) than group 2 (11 years) / exposure of mothers to radiation for different periods of time ;</li> <li>6 <i>idea that</i> exposure to radioactivity, calculated / measured, differently for the two groups ;</li> <li>7 <i>idea that</i> types of radiation might be different in the two groups ;</li> <li>8 <i>idea that</i> records have been lost ;</li> <li>9 <i>idea that</i> method of record keeping has been changed ;</li> <li>10 <i>idea that</i>, people who moved away were not followed up / large number of outcomes not known ;</li> <li>11 <i>idea that</i> there may be other factors contributing to blood cancers that were not taken into account ;</li> </ol> | 3     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                     | Marks    |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(d)     | <p><b>three of:</b></p> <p>seeds ; ;<br/> e.g. same / similar, number of seeds<br/> same, species / type / variety, of plant<br/> same age of seeds / storage of seeds</p> <p>stated conditions for germination ; ;<br/> e.g. light<br/> water availability<br/> temperature<br/> oxygen</p> <p>same, source / type, of radiation ;</p> <p>same background radiation ;</p> | <b>3</b> |